Do Now · retrieval

Lesson 064 · Thinking Concurrently: Independent Work

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a process dependency?

Q2

What is throughput?

Q3

What is a race condition?

Q4

A rising-edge D-type flip-flop starts with Q=0. At a rising edge D=1, so Q becomes 1. Immediately afterwards D becomes 0. Explain why Q stays 1 through the following falling edge.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify — Find tasks that do not depend on each other's intermediate results. Independent work can overlap.

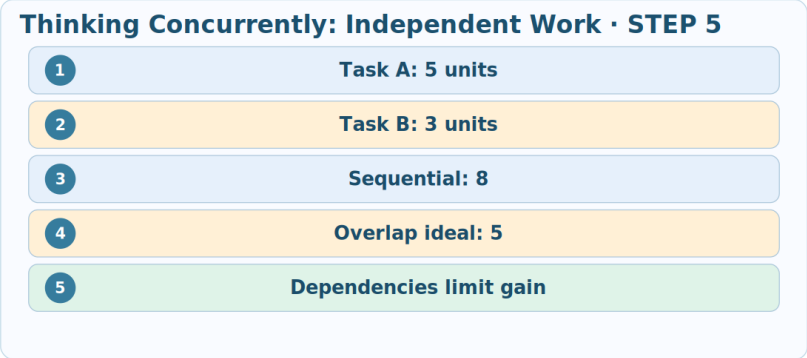
Core idea 2

Model — Distinguish concurrency (progress interleaved) from parallelism (truly simultaneous hardware execution).

Core idea 3

Evaluate — Evaluate scheduling overhead and shared-resource/race risks, and explain why dependencies limit possible speedup.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

concurrency — Multiple tasks make progress in overlapping period.

parallelism — Actual simultaneous execution.

critical path — Longest dependency chain.

race condition — Incorrect timing-dependent shared state.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

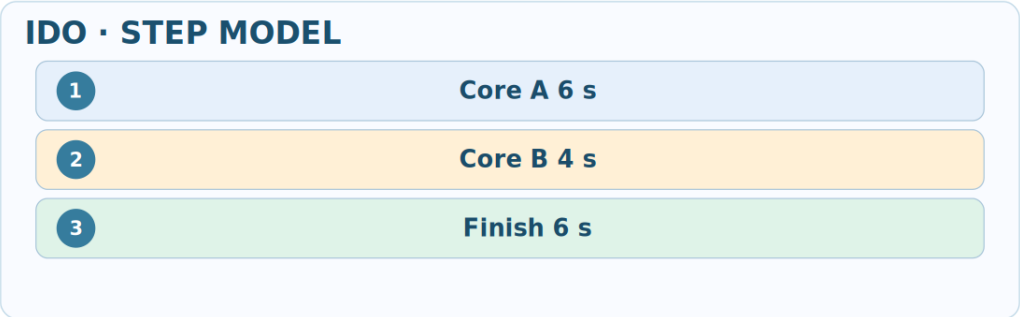
Problem

Calculate minimum ideal time for independent image operations 6s and 4s on separate cores.

Follow the worked method

1. Explain the decision — Only independent tasks can safely overlap without waiting for each other.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the decision
Extra processors do not remove logical dependency.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Independent jobs take 7s and 4s. Calculate sequential and ideal parallel completion times. [4]

Question 2

Give two potentially concurrent stages of a school app and two stages that must be sequential. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Two threads increment shared counter via read, add, write and occasionally lose updates. Explain race and one prevention. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Doubling the number of CPU cores always halves task completion time.

Use key vocabulary from memory

1. Explain concurrency in your own words.

2. Explain parallelism in your own words.

3. Explain critical path in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.